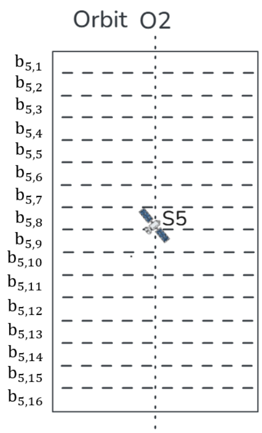

Fig. 1. Illustrative satellite-fixed beam layout with 16 downlink beams indexed from north to south.

IV. Method

A. Method Overview

Chapter III formulates the service recovery problem with user association, beam power, satellite power budget, and EPFD constraints. Under the EPFD constraint, beams from one or more LEO satellites may need to be shut down. A satellite is regarded as critical when at least one of its beams covers the GS and must be shut down to satisfy the EPFD constraint. It should be noted that a critical satellite is not a fixed satellite identity. Instead, this role is assigned according to the satellite geometry and EPFD condition in each time slot. A satellite is not predefined as critical before entering the near-inline period. It is regarded as critical only during the time slots in which its beams must be shut down due to the GS geometry and the EPFD constraint. Once the satellite moves away and its beams no longer need to be shut down for EPFD compliance, the satellite is no longer treated as a critical satellite. Since this criterion is evaluated for all visible LEO satellites, multiple critical satellites may exist in the same time slot. To recover the service of users affected by beam shutdown on critical satellites, this chapter presents an EPFD-aware beam reassociation framework for service recovery.

The proposed framework is designed based on predictable satellite motion, beam footprints, EPFD contributions, and user service requirements. For each time slot, the framework determines which beams need to be shut down to satisfy the EPFD constraint and which neighboring satellites can support service recovery according to the current satellite geometry and EPFD condition. It then constructs candidate beam relation graphs to describe the possible reassociation relations between beams of critical satellites and neighboring helper satellites.

To explain the beam role identification and beam reassociation relations introduced later, this chapter uses a simplified OneWeb-like multi-beam scenario [24] for illustration. As shown in Fig. 1, each satellite employs 16 satellite-fixed downlink beams, which are indexed from beam 1 to beam 16 according to their ground footprints from north to south. All satellites use the same beam layout and indexing rule, so each beam can be uniquely identified by its satellite and beam index.

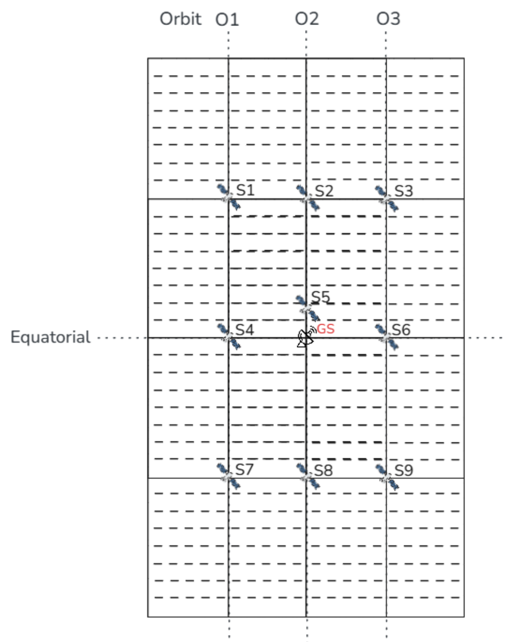

Fig. 2. Illustrative OneWeb-like local satellite geometry across three neighboring orbital planes. Each satellite employs the 16-beam layout.

As shown in Fig. 2, a local OneWeb-like satellite geometry across three neighboring orbital planes, denoted by P1, P2, and P3, is used to illustrate the multi-satellite coverage relationship. Satellites S1 to S9 represent satellites at different orbital positions, while the GS is located near the equator. Each satellite employs the same 16-beam layout shown in Fig. 1. Because the service regions of neighboring satellites partially overlap, the same ground area may be covered by beams from different satellites. These overlapping coverage relations provide alternative serving-beam candidates for user reassociation and motivate the proposed EABR framework.

The proposed framework consists of three major online procedures. First, Safe-Beam Reassociation for Helper Users (SBR) uses the residual capacity of critical safe beams to reassociate part of the helper users originally served by helper release beams. This reduces the loading of helper release beams and releases part of the helper satellite transmit power. Second, EPFD-constrained power allocation assigns the available helper satellite power to helper recovery beams according to their priority-weighted recovery demand. Third, Helper-Beam Reassociation for Closed-Beam Users (HBR) uses helper recovery beams to recover closed-beam users. Through these procedures, the proposed framework aims to improve service continuity for closed-beam users while maintaining EPFD compliance and power constraints.

In summary, the proposed EPFD-aware beam reassociation framework consists of the following steps:

- **Time-slot beam role record:** Precompute the critical satellites, closed beams, critical safe beams, helper release beam candidates, helper recovery beam candidates, and candidate beam relation graphs for each time slot.
- **Safe-Beam Reassociation for Helper Users (SBR):** Re-associate selected helper users from helper release beams

to critical safe beams, reducing helper beam loading and releasing helper satellite transmit power.

- **EPFD-Constrained Power Allocation for Helper Recovery Beams:** Allocate the available helper satellite power to helper recovery beams according to the priority-weighted recovery demand, subject to beam power and aggregate EPFD constraints.
- **Helper-Beam Reassociation for Closed-Beam Users (HBR):** Reassociate selected closed-beam users to helper recovery beams and recover their service under the available power of helper recovery beams.

B. Time-Slot Beam Role Record

Before the online reassociation procedures are executed, the system constructs a time-slot beam role record for each time slot. This record stores the critical satellites, neighboring helper satellites, and the beam pairs that may be used for reassociation in the current time slot. These records are mainly built according to predictable satellite geometry, beam footprints, and EPFD contributions.

1) Critical Satellite and Beam Role Identification: For each time slot t, the system first checks whether the aggregate EPFD generated by all active beams of visible LEO satellites satisfies the EPFD limit. If the aggregate EPFD is already below the limit, beam shutdown is not required in this time slot, and the critical satellite set is empty. If the aggregate EPFD exceeds the limit, the system examines the beam-level EPFD contribution of each active beam on all visible LEO satellites and sorts the beams in descending order according to $\mathrm{EPFD}_{s,b}(t)$. The system then sequentially shuts down beams with higher EPFD contributions and updates the aggregate EPFD after each beam shutdown until the aggregate EPFD satisfies the EPFD limit.

Let $\mathcal{B}_s^{\mathrm{off}}(t)$ denote the set of beams shut down on satellite s in time slot t. After the beam shutdown process is completed, all satellites with a non-empty $\mathcal{B}_s^{\mathrm{off}}(t)$ form the critical satellite set, denoted by $\mathcal{S}^{\mathrm{C}}(t)$.

For each critical satellite s, the beams that are not shut down and remain active form the safe-beam set $\mathcal{B}_s^{\mathrm{safe}}(t)$, and are referred to as safe beams. An individual closed beam and safe beam are denoted by $b_{s,l}^{\mathrm{off}}$ and $b_{s,l}^{\mathrm{safe}}$, respectively. Therefore, the critical satellite set is determined by the beam shutdown result of each time slot. If beams from different satellites must be shut down to satisfy the EPFD constraint, these satellites are included in $\mathcal{S}^{\mathrm{C}}(t)$ at the same time.

2) Helper Satellite and Beam Role Identification: After critical set identification is completed, the system constructs the helper satellite set, denoted by $\mathcal{S}^{\mathrm{H}}(t)$. Similar to the critical satellite role, the helper satellite is also a time-dependent operational role rather than a permanent satellite identity. A neighboring satellite is regarded as a helper only when its beams have feasible overlap relations with the current critical satellites and can support the service recovery process in the current time slot. Once the corresponding critical satellites no longer need to shut down beams for EPFD compliance, the helper relation is also released. For each helper satellite candidate, the system checks which of its beams overlap with the beams of the current critical satellites. Based on these overlap relations, each helper beam is then classified as a possible helper release beam or helper recovery beam.

If the service area of a helper beam overlaps with the footprint of critical safe beams, this helper beam is marked as a helper release beam candidate, denoted by $b_{h,k}^{\mathrm{rls}}$. The set of helper release beam candidates on helper satellite h at time slot t is denoted by $\mathcal{B}_h^{\mathrm{rls}}(t)$. A helper release beam candidate represents a helper beam whose served users overlap with critical safe beams and whose loading may be reduced through reassociation.

If the service area of a helper beam overlaps with the footprint of critical closed beams, this helper beam is marked as a helper recovery beam candidate, denoted by $b_{h,k}^{\mathrm{rec}}$. The set of helper recovery beam candidates on helper satellite h at time slot t is denoted by $\mathcal{B}_h^{\mathrm{rec}}(t)$. A helper recovery beam candidate represents a helper beam whose coverage can include users affected by closed beams.

If a helper satellite h has a non-empty $\mathcal{B}_h^{\mathrm{rls}}(t)$ or a non-empty $\mathcal{B}_h^{\mathrm{rec}}(t)$, the satellite is included in the helper satellite set $\mathcal{S}^{\mathrm{H}}(t)$.

3) Candidate Beam Relation Graph Construction: After the critical satellites, helper satellites, and beam roles are identified, the system constructs the candidate beam relation graphs according to the beam shutdown results and beam footprint overlap in each time slot. Fig. 3 shows the critical satellites, helper satellites, and closed-beam results in the illustrative scenario. If a critical safe beam overlaps with a helper release beam candidate, the corresponding beam pair is recorded in the safe-release candidate graph. If a critical closed beam overlaps with a helper recovery beam candidate, the corresponding beam pair is recorded in the closed-recovery candidate graph.

As shown in Fig. 4, the safe-release candidate graph is constructed from the overlap relations between critical safe beams and helper release beam candidates in the illustrative geometry. This graph records which critical safe beams have footprint overlap with helper release beam candidates. It provides the candidate beam pairs used in Safe-Beam Reassociation for Helper Users (SBR), and the detailed reassociation procedure is described in Section IV-C.

As shown in Fig. 5, the closed-recovery candidate graph is constructed from the overlap relations between critical closed beams and helper recovery beam candidates in the illustrative geometry. Each edge records a beam pair with footprint overlap. This graph provides the candidate beam pairs used in Helper-Beam Reassociation for Closed-Beam Users (HBR), and the detailed reassociation procedure is described in Section IV-E.

It should be noted that the candidate beam relation graphs are used only to illustrate the footprint overlap relations between beams. They show which beam pairs may have overlap in the current time slot, but they do not directly determine the reassociation result. The actual reassociation decisions are made later in Safe-Beam Reassociation for Helper Users (SBR) and Helper-Beam Reassociation for Closed-Beam Users (HBR).

C. Safe-Beam Reassociation for Helper Users

Safe-Beam Reassociation for Helper Users (SBR) is the first reassociation procedure in the proposed framework. It is performed based on the safe-release candidate graph constructed

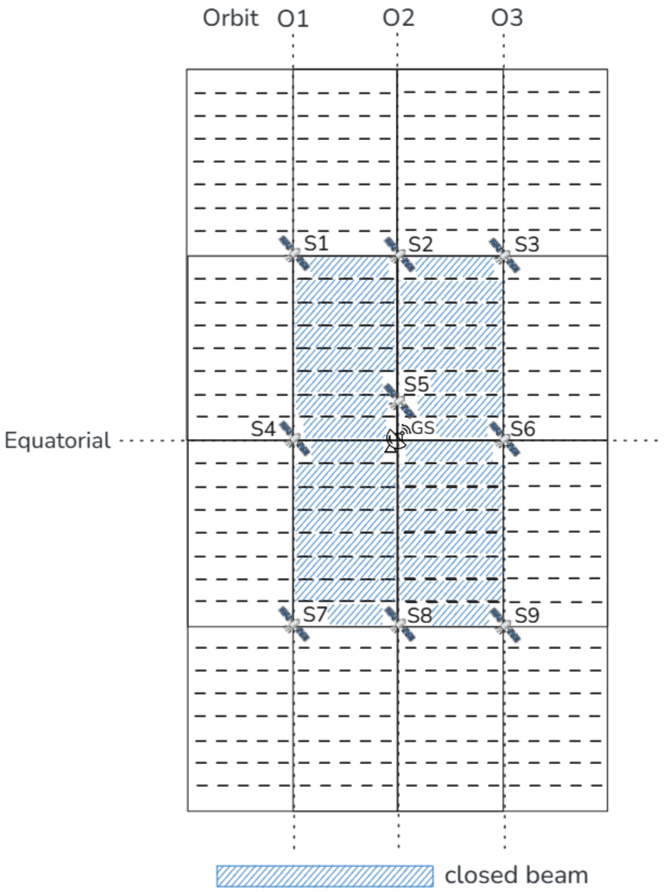

Fig. 3. Illustrative beam-role identification result in the OneWeb-like local geometry. Satellites S2, S5, and S8 are critical satellites, while S1, S3, S4, S6, S7, and S9 are helper satellites. The hatched regions indicate closed beams on the critical satellites.

in the previous subsection. The purpose of SBR is to let critical safe beams take over selected users originally served by helper release beams. After these users are reassociated to critical safe beams, the corresponding helper release beams can reduce their original service load. The released helper-side resources are then returned to the power budget of the helper satellite.

As shown in Fig. 4, the left-side nodes are critical safe beams, and the right-side nodes are helper release beam candidates. For example, the critical safe beam $b_{2,1}^{\mathrm{safe}}$ is connected to two helper release beam candidates, $b_{1,1}^{\mathrm{rls}}$ and $b_{3,1}^{\mathrm{rls}}$. This means that safe beam 1 of critical satellite S2 overlaps with release beam 1 of helper satellites S1 and S3. These beam pairs can be used as candidate reassociation choices in SBR.

1) Candidate Helper User Set Construction: For a helper release beam candidate, the users currently served by this beam are called helper-side users. For an edge e, not all helper-side users are relevant to SBR. The helper-side users that are also covered by the critical safe beam connected to the same edge can be considered for reassociation through this edge. These users form the candidate helper-user set of edge e, denoted by $\mathcal{U}_e^{\mathrm{cand}}(t)$. At this stage, $\mathcal{U}_e^{\mathrm{cand}}(t)$ only represents the users that may be considered for reassociation through edge e, and no user is reassociated yet.

2) Release Score Evaluation: For each edge e, SBR evaluates

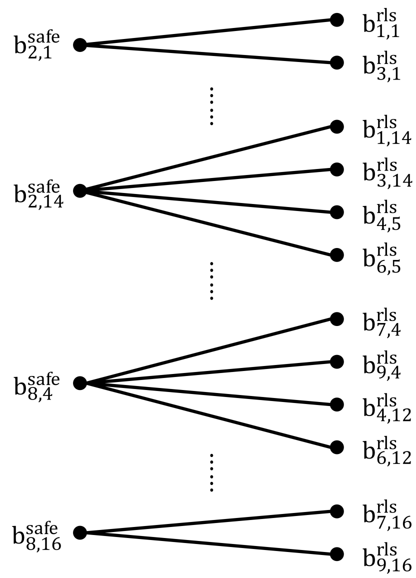

Fig. 4. Safe-release candidate graph between critical safe beams and helper release beam candidates. The left-side nodes $b_{s,\ell}^{\mathrm{safe}}$ denote safe beams on critical satellites, and the right-side nodes $b_{h,k}^{\mathrm{rls}}$ denote helper release beam candidates. An edge $e = (b_{s,\ell}^{\mathrm{safe}}, b_{h,k}^{\mathrm{rls}})$ exists when the two beams have footprint overlap.

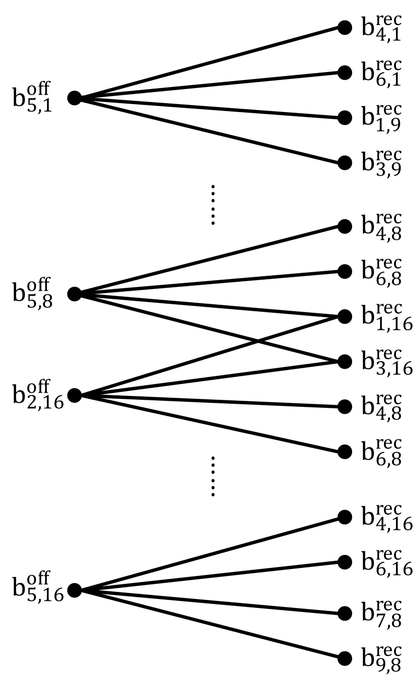

Fig. 5. Closed-recovery candidate graph between critical closed beams and helper recovery beam candidates. The left-side nodes $b_{s,\ell}^{\mathrm{off}}$ denote closed beams on critical satellites, and the right-side nodes $b_{h,k}^{\mathrm{rec}}$ denote helper recovery beam candidates. An edge $e = (b_{s,\ell}^{\mathrm{off}}, b_{h,k}^{\mathrm{rec}})$ exists when the two beams have footprint overlap.

how much helper-side service demand can be moved from the helper release beam to the connected critical safe beam. The evaluation is performed over the candidate helper-user set $\mathcal{U}_e^{\mathrm{cand}}(t)$. From this set, SBR searches for a feasible user subset $\mathcal{U}_e^{\mathrm{sel}}(t)$ that can be taken over by the critical safe beam.

A candidate helper user can be included in $\mathcal{U}_e^{\mathrm{sel}}(t)$ only when two conditions are satisfied. First, after this user is moved from the helper release beam to the connected critical safe beam, its service satisfaction must not be lower than its original satisfaction on the helper release beam. If the user would receive lower satisfaction after reassociation, this user remains served by the original helper release beam and is not selected for this edge. Second, the critical safe beam must still have enough remaining capacity to serve the selected users. The total demand of the selected users in $\mathcal{U}_e^{\mathrm{sel}}(t)$ cannot be larger than the remaining capacity of the critical safe beam.

The release score of edge e is defined as the maximum total service demand that can be moved through this edge:

$$\Phi_e^{\mathrm{SBR}}(t) = \max_{\mathcal{U}_e^{\mathrm{sel}}(t) \subseteq \mathcal{U}_e^{\mathrm{cand}}(t)} \sum_{u \in \mathcal{U}_e^{\mathrm{sel}}(t)} D_u, \tag{10}$$

where $\mathcal{U}_e^{\mathrm{sel}}(t)$ is restricted to the feasible user subsets that satisfy the two conditions described above. A larger $\Phi_e^{\mathrm{SBR}}(t)$ means that more helper-side service demand can be moved away from the helper release beam through edge e. Therefore, this score is designed to prioritize the edge that can release the largest amount of helper-side service load while satisfying the capacity and non-degradation conditions. If no feasible user subset can be found, or if the connected critical safe beam has no remaining capacity, the release score of this edge is set to zero.

3) Iterative Edge Selection and Update: The safe-release candidate graph may contain several connected components. Each component includes the critical safe beams and helper release beam candidates that are connected through footprint overlap. SBR processes these components iteratively. In each component, SBR selects the edge with the highest release score. The selected edge is marked as an active safe-release relation, and the feasible user subset corresponding to this selected edge is reassociated from the helper release beam to the connected critical safe beam. After the reassociation, the loading and remaining capacity of the critical safe beam are updated, and the service load originally carried by the helper release beam is reduced. The released helper-side resources are added back to the power budget of the helper satellite.

As illustrated in Fig. 6, the selected edge is marked in blue with a check mark. After one edge is selected, the release scores of the remaining edges in the same component are updated because the remaining capacity of the connected critical safe beam may have changed. If a critical safe beam no longer has enough remaining capacity to take over additional helper users, the remaining edges connected to this safe beam are assigned a release score of zero. SBR then selects the next edge with the highest release score in the component. This process continues until the remaining edges in each component have zero release scores. At the end of SBR, the graph contains the selected safe-release relations and the remaining zero-score edges. The corresponding helper release beams reduce their original service load, and the released helper-side resources are added back to the power budget of the helper satellite.

D. EPFD-Constrained Power Allocation for Helper Recovery Beams

After SBR is completed, the transmit power released by helper release beams is added back to the available power budget of the corresponding helper satellite. This available power budget consists of the residual satellite power remaining after the default beam-power allocation and the transmit power released through SBR. The system then allocates the available power to helper recovery beams on the same helper satellite for serving closed-beam users.

To determine the power allocation order, the system first checks the candidate users connected to each helper recovery beam according to the closed-recovery candidate graph. These candidate users come from critical closed beams that have footprint overlap with the helper recovery beam. The system then calculates the priority-weighted recovery demand of each helper recovery beam based on the service priorities and traffic demands of these candidate users. If a helper recovery beam is connected to more high-priority users or users with larger traffic demand, the beam has a higher recovery demand value.

After the recovery demand value is calculated, the helper recovery beams are sorted in descending order of priority-weighted recovery demand. The available power budget of the helper satellite is then allocated to the helper recovery beams according to this order. For each helper recovery beam, the amount of additional power that can be allocated is determined under the beam power upper bound, the available power budget of the helper satellite, and the aggregate EPFD constraint. Before this power allocation is accepted, the aggregate EPFD is recalculated using the updated transmit power of the current helper recovery beam. If the resulting aggregate EPFD remains below the applicable EPFD limit, the power allocation is accepted. Otherwise, the allocated power is reduced to the maximum feasible amount that still satisfies the EPFD constraint. If no additional power can be allocated to the current helper recovery beam, its transmit power remains unchanged, and the allocation procedure proceeds to the next helper recovery beam. After each accepted allocation, the available power budget of the helper satellite and the aggregate EPFD are updated. This process continues until the available power budget is exhausted or no further feasible power allocation can be made under the beam power and EPFD constraints.

E. Helper-Beam Reassociation for Closed-Beam Users

Helper-Beam Reassociation for Closed-Beam Users (HBR) is performed after helper recovery beams obtain available recovery resources. It is based on the closed-recovery candidate graph constructed in Section IV-B. The purpose of HBR is to reassociate users originally served by critical closed beams to neighboring helper recovery beams. Through this step, users affected by beam shutdown can receive service from helper satellites.

As shown in Fig. 5, the left-side nodes are critical closed beams, and the right-side nodes are helper recovery beam candidates. For example, the closed beam $b_{5,1}^{\mathrm{off}}$ is connected to several helper recovery beam candidates, such as $b_{4,1}^{\mathrm{rec}}$, $b_{6,1}^{\mathrm{rec}}$, $b_{1,9}^{\mathrm{rec}}$, and $b_{3,9}^{\mathrm{rec}}$. This means that users originally located under closed beam 1 of critical satellite S5 may also be covered by these helper recovery beams. These beam pairs can be used as candidate reassociation choices in HBR.

1) Candidate Closed-Beam User Set Construction: For a critical closed beam, the users originally served by this beam before shutdown are called closed-beam users. These users lose their original serving beam after the beam is shut down and therefore need to be reassociated to helper recovery beams. For an edge e in the closed-recovery candidate graph, not all closed-beam users can be served through this edge. The closed-beam users that are also covered by the helper recovery beam connected to the same edge can be considered for reassociation. These users form the candidate closed-beam user set of edge e, denoted by $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$. At this stage, $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$ only represents the users that may be recovered through edge e, and no user is reassociated yet.

2) Recovery Score Evaluation: For each edge e, HBR evaluates how much service can be recovered if some candidate closed-beam users are reassociated to the connected helper recovery beam. The evaluation is performed over the candidate closed-beam user set $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$. From this set, HBR searches for a feasible user subset $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ that can be served by the helper recovery beam under its available recovery resources.

A candidate closed-beam user can be included in $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ only when the helper recovery beam still has enough available capacity to serve the selected users. Since the available recovery capacity of a helper recovery beam is limited, HBR first favors high-priority users and then selects the user subset that can achieve the largest recovered satisfaction under the available capacity. Therefore, the recovery score considers both the user priority and the estimated satisfaction after reassociation. The recovery score of edge e is defined as

$$\Phi_e^{\mathrm{HBR}}(t) = \max_{\mathcal{U}_e^{\mathrm{rec,sel}}(t) \subseteq \mathcal{U}_e^{\mathrm{rec,cand}}(t)} \sum_{u \in \mathcal{U}_e^{\mathrm{rec,sel}}(t)} w_{\pi(u)}\, \hat{s}_{u,e}^{\mathrm{rec}}(t), \tag{11}$$

where $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ is restricted to the feasible user subsets that can be supported by the connected helper recovery beam. Here, $w_{\pi(u)}$ is the priority weight of user u, and $\hat{s}_{u,e}^{\mathrm{rec}}(t)$ is the estimated satisfaction of user u if it is reassociated through edge e. A larger $\Phi_e^{\mathrm{HBR}}(t)$ means that this edge can recover a user subset with higher priority-weighted satisfaction. If no candidate closed-beam user can be served through this edge, or if the connected helper recovery beam has no available recovery capacity, the recovery score of this edge is set to zero.

3) Iterative Edge Selection and Update: The closed-recovery candidate graph may also contain several connected components. Each component includes the critical closed beams and helper recovery beam candidates that are connected through footprint overlap. HBR processes these components iteratively. In each component, HBR selects the edge with the highest recovery score. The selected edge is marked as an active closed-recovery relation, and the corresponding feasible user subset is reassociated from the critical closed beam to the connected helper recovery beam. After the reassociation, the remaining users under the critical closed beam are updated, and the remaining recovery capacity of the helper recovery beam is also updated.

As illustrated in Fig. 7, the selected edge is marked in blue with a check mark. After one edge is selected, the recovery scores of the remaining edges in the same component are updated because the remaining users and the available recovery capacity may have changed. If a critical closed beam has no remaining user to recover, the remaining edges connected to this closed beam are assigned a recovery score of zero. Similarly, if a helper recovery beam has no remaining recovery capacity, the related edges are also assigned a recovery score of zero. HBR then selects the next edge with the highest recovery score in the component. This process continues until the remaining edges in each component have zero recovery scores. At the end of HBR, the framework obtains the selected closed-recovery relations and the closed-beam users that are moved to helper recovery beams. These users can then receive service from neighboring helper satellites.

F. Overall Procedure and State Update

This section summarizes the overall procedure of EPFD-Aware Beam Reassociation for Service Recovery. The previous subsections describe the time-slot beam role record, SBR, EPFD-constrained power allocation, and HBR. This section connects these steps into a complete time-slot procedure and explains the system state that must be updated after each time slot.

Before the online reassociation procedures are executed, the system precomputes the time-slot beam role records for all time slots according to predictable satellite motion, beam footprints, user coverage, and beam-level EPFD contributions. If the aggregate EPFD of a time slot already satisfies the EPFD limit, beam shutdown is not required in that time slot. If the aggregate EPFD exceeds the limit, the corresponding critical satellites, closed beams, critical safe beams, helper release beam candidates, and helper recovery beam candidates are stored in the record.

The system further constructs the safe-release candidate graphs and the closed-recovery candidate graphs. The safe-release candidate graph is used as the input of SBR to determine which helper-side users may be reassociated to critical safe beams, thereby releasing helper satellite transmit power. The closed-recovery candidate graph is used as the input of HBR to determine which closed-beam users may be reassociated to helper recovery beams.

In each time slot, the system retrieves the corresponding time-slot beam role record. If no closed beam exists in the time slot, the system does not need to execute the following reassociation procedures. If closed beams exist, the system first performs SBR. SBR selects active safe-release relations according to the release score of each safe-release edge and reassociates the feasible helper-side user subsets to the corresponding critical safe beams. After SBR is completed, the loading of helper release beams is reduced, and the corresponding released transmit power is added to the available power budget of the helper satellite.

Before HBR is executed, the system performs EPFD-constrained power allocation. This step allocates the available power of the helper satellite to helper recovery beams according to their priority-weighted recovery demand, subject to beam power and aggregate EPFD constraints. After the allocation, each helper recovery beam obtains updated available recovery power.

Finally, the system performs HBR. HBR selects active closed-recovery relations according to the recovery score of each closed-recovery edge and reassociates the feasible closed-beam user subsets to the corresponding helper recovery beams.

Algorithm 1 Overall Procedure of EPFD-Aware Beam Reassociation for Service Recovery

Input: LEO constellation and beam configuration, GSO–GS and EPFD model, user distribution with service demand and priority weights, and beam power configuration
Output: User association sequence $\mathbf{X}$ and beam power allocation sequence $\mathbf{P}$

Initialize $\mathbf{X}$ and $\mathbf{P}$
Precompute the time-slot beam role records for all time slots
Construct the safe-release candidate graphs for all time slots
Construct the closed-recovery candidate graphs for all time slots
for each time slot t **do**
  Retrieve the time-slot beam role record at time slot t
  if no closed beam exists **then**
    continue
  end if
  // Safe-Beam Reassociation for Helper Users
  Initialize the active safe-release relation set as empty
  Evaluate $\Phi_e^{\mathrm{SBR}}(t)$ of each safe-release edge e by (10)
  while there exists a safe-release edge with $\Phi_e^{\mathrm{SBR}}(t) > 0$ **do**
    Divide the current safe-release candidate graph into connected components
    for each connected component **do**
      Select the edge e with the maximum $\Phi_e^{\mathrm{SBR}}(t)$
      Reassociate the selected user subset $\mathcal{U}_e^{\mathrm{sel}}(t)$ of edge e to the corresponding critical safe beam
      Add edge e to the active safe-release relation set
      Mark edge e as selected
    end for
    Update the safe-release candidate graph and recompute the release scores by (10)
  end while
  // EPFD-Constrained Power Allocation for Helper Recovery Beams
  Update the available power budget of each helper satellite by adding the transmit power released through SBR
  Compute the priority-weighted recovery demand of each helper recovery beam
  Sort helper recovery beams in descending order of priority-weighted recovery demand
  Allocate the available helper satellite power to helper recovery beams subject to beam power and aggregate EPFD constraints
  Recalculate and update the aggregate EPFD after each accepted power allocation
  Update the available power of helper recovery beams
  // Helper-Beam Reassociation for Closed-Beam Users
  Initialize the active closed-recovery relation set as empty
  Evaluate $\Phi_e^{\mathrm{HBR}}(t)$ of each closed-recovery edge e by (11)
  while there exists a closed-recovery edge with $\Phi_e^{\mathrm{HBR}}(t) > 0$ **do**
    Divide the current closed-recovery candidate graph into connected components
    for each connected component **do**
      Select the edge e with the maximum $\Phi_e^{\mathrm{HBR}}(t)$
      Reassociate the selected user subset $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ of edge e to the corresponding helper recovery beam
      Add edge e to the active closed-recovery relation set
      Mark edge e as selected
    end for
    Update the closed-recovery candidate graph and recompute the recovery scores by (11)
  end while
  Update $\mathbf{X}(t)$ and $\mathbf{P}(t)$
end for
return $\mathbf{X}$, $\mathbf{P}$

References

[1] M. Giordani and M. Zorzi, "Non-terrestrial networks in the 6g era: Challenges and opportunities," *IEEE Network*, vol. 35, no. 2, pp. 244–251, 2021.

[2] X. Lin, S. Cioni, G. Charbit, R. Chuberre, S. Hellsten, and J.-F. Boutillon, "On the path to 6g: Embracing the next wave of low earth orbit satellite access," *IEEE Communications Magazine*, vol. 59, no. 5, pp. 36–42, 2021.

[3] E. Juan, M. Lauridsen, J. Wigard, and P. Mogensen, "Handover solutions for 5g low-earth orbit satellite networks," *IEEE Access*, vol. 10, pp. 93 309–93 325, 2022.

[4] M. Al-Ansi, J. Querol, M. Alsenwi, E. Lagunas, and S. Chatzinotas, "Enhancing handover performance in LEO satellite networks with multi-connectivity and conditional handover approach," in *Proceedings of the International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*, 2024.

[5] A. Gonzalez-Garrido, J. Querol, H. Wymeersch, and S. Chatzinotas, "Joint communication and navigation from LEO multi-beam satellite," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 4384–4404, 2025.

[6] D. Christopoulos, S. Chatzinotas, J. Krause, and B. Ottersten, "Multi-user detection in multibeam mobile satellite systems: A fair performance evaluation," *IEEE Transactions on Communications*, vol. 61, no. 12, pp. 4963–4974, 2013.

[7] X. Zhu and C. Jiang, "Integrated satellite-terrestrial networks toward 6g: Architectures, applications, and challenges," *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 437–461, 2022.

[8] O. B. Al-Hraishawi, H. Chaaban, S. Kisseleff, E. Lagunas, and S. Chatzinotas, "A survey on non-geostationary satellite systems: The communication perspective," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 101–132, 2023.

[9] ITU-R, "Radio regulations," International Telecommunication Union, Geneva, Switzerland, 2020, see Article 22 for NGSO–GSO interference limits and EPFD framework.

[10] ——, "Functional description to be used in developing software tools for determining conformity of non-geostationary-satellite orbit fixed-satellite service systems or networks with limits contained in article 22 of the radio regulations," International Telecommunication Union Radiocommunication Sector (ITU-R), Geneva, Switzerland, Recommendation S.1503-3, 2018.

[11] International Telecommunication Union Radiocommunication Sector (ITU-R), "Satellite antenna radiation patterns for non-geostationary orbit satellite antennas operating in the fixed-satellite service below 30 ghz," ITU-R, Geneva, Switzerland, Recommendation S.1528, 2001.

[12] ITU-R, "Reference fss earth-station radiation patterns for use in interference assessment involving non-gso satellites in frequency bands between 10.7 ghz and 30 ghz," ITU-R, Geneva, Switzerland, Recommendation S.1428-1, 2001.

[13] ——, "Calculation of free-space attenuation," International Telecommunication Union Radiocommunication Sector, Geneva, Switzerland, Recommendation P.525.

[14] ——, "Attenuation by atmospheric gases and related effects," International Telecommunication Union Radiocommunication Sector, Geneva, Switzerland, Recommendation P.676.

[15] ETSI, "5g; system architecture for the 5g system (5gs) (3gpp ts 23.501 version 16.14.0 release 16)," European Telecommunications Standards Institute, ETSI TS 123 501 V16.14.0, Sep. 2022, 3GPP TS 23.501 version 16.14.0 Release 16.

[16] C. Braun, A. M. Voicu, L. Simić, and P. Mähönen, "Should we worry about interference in emerging dense NGSO satellite constellations?" in *Proc. IEEE Int. Symp. Dyn. Spectr. Access Netw. (DySPAN)*, 2019, pp. 1–10.

[17] A. Hills, J. M. Peha, and J. Munk, "Feasibility of using beam steering to mitigate ku-band leo-to-geo interference," *IEEE Access*, vol. 10, pp. 74 023–74 032, 2022.

[18] M. Jalali, F. Ortiz, E. Lagunas, S. Kisseleff, L. Emiliani, and S. Chatzinotas, "Joint power and tilt control in satellite constellation for NGSO–GSO interference mitigation," *IEEE Open Journal of Vehicular Technology*, vol. 4, pp. 545–557, 2023.

[19] T. Li, J. Jin, W. Li, Z. Ren, and L. Kuang, "Research on interference avoidance effect of oneweb satellite constellation's progressive pitch strategy," *International Journal of Satellite Communications and Networking*, vol. 39, no. 5, pp. 524–538, 2021.

[20] M. Jalali, E. Lagunas, A. Haqiqatnejad, S. Kisseleff, and S. Chatzinotas, "Downlink beamforming strategies for interference-aware NGSO satellite systems," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 3468–3483, 2024.

[21] G. Cocco, M. Angelone, and A. I. Pérez-Neira, "Co-channel interference cancellation at the user terminal in multibeam satellite systems," in *Proceedings of the IEEE International Conference on Communications (ICC)*, 2014, pp. 3992–3997.

[22] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[23] J. Lei and M. A. Vázquez-Castro, "Multibeam satellite frequency/time duality study and capacity optimization," *Journal of Communications and Networks*, vol. 13, no. 5, pp. 472–480, 2011.

[24] WorldVu Satellites Limited, "Oneweb non-geostationary satellite system," FCC IBFS File No. SAT-LOI-20160428-00041, Technical Attachment, Washington, DC, USA, 2016, oneWeb NGSO system description.

[25] Federal Communications Commission, "Space Exploration Holdings, LLC, Request for Modification of the Authorization for the SpaceX NGSO Satellite System," Order and Authorization and Order on Reconsideration, FCC 21-48, Washington, DC, USA, Apr. 2021.

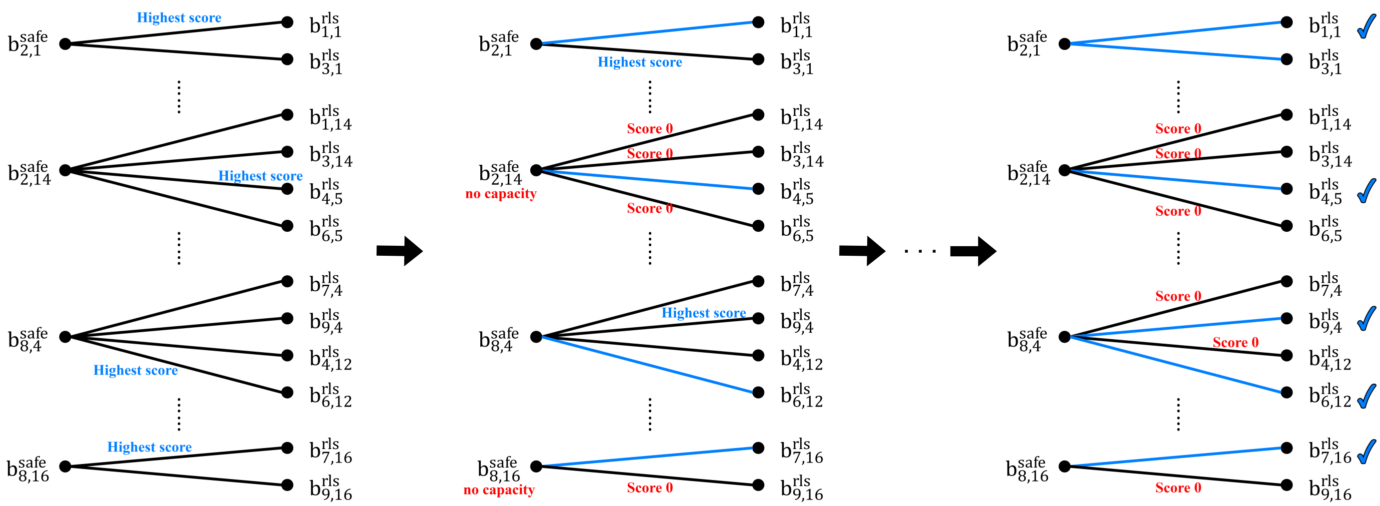

Fig. 6. Example of iterative edge activation and graph update in SBR.

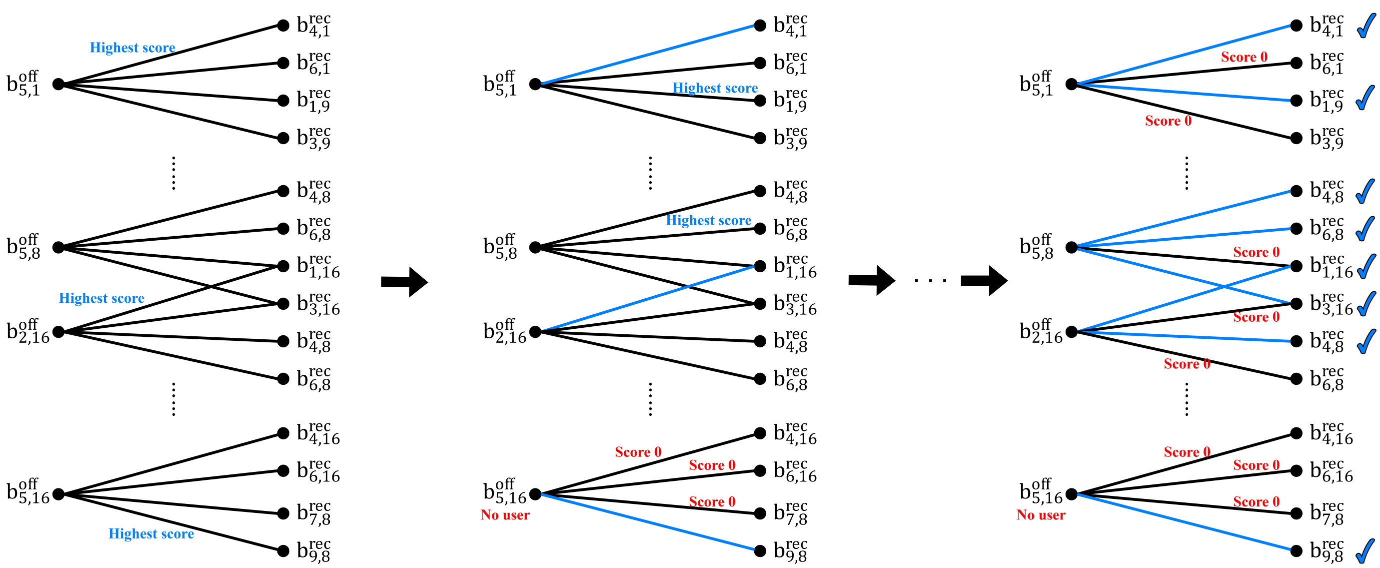

Fig. 7. Example of iterative edge activation and graph update in HBR.

[26] ——, "Space Station Applications Accepted for Filing," Public Notice, Report No. SAT-01632, Washington, DC, USA, May 2022.

[27] Analytical Graphics, Inc., "Systems tool kit (stk)," https://www.agi.com/products/stk, 2023, online; accessed 21-Jul-2025.

[28] The MathWorks, Inc., "Matlab r2023b," https://www.mathworks.com/products/matlab.html, 2023, online; accessed 21-Jul-2025.

[29] ITU-R, "Interference mitigation techniques and frequency sharing in the bands 37.5–42.5 ghz and 47.2–50.2 ghz between geostationary-satellite fixed-satellite service networks and non-geostationary-satellite fixed-satellite service systems," ITU-R, Geneva, Switzerland, Recommendation S.1655-0, 2003.